

# The Loyola Index Family: What a Degree-Imbalance Coordinate Can and Cannot Resolve

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## Abstract

The two-parameter Gutman–Milovanović family  $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$  unifies the Zagreb, Randić, sum-connectivity and related degree-based indices. We study its normalized degree-imbalance exponential extension, the *Loyola index family*

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right),$$

with  $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$  identically. We ask what the imbalance coordinate  $\gamma$  adds to the parent family. On a ratio plane that separates size from shape, the pure- $\gamma$  points distinguish two graphs exactly when their histograms of edge imbalances differ. On small trees this is the discrimination limit of all degree-based indices, and we prove that the limit is lost at larger orders. We bound the number of times two graphs exchange order along  $\gamma$  and characterise the directions in parameter space that a given graph can identify. Pre-registered tests on alkane properties show that  $\gamma$  gives no predictive gain over the parent family.

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# 1 Introduction

Degree-based topological indices form one of the most developed descriptor classes of chemical graph theory. The class runs from the Randić and Zagreb indices [15, 29] and their parametric extensions, the general Randić and sum-connectivity families [5, 37], to the Sombor family and its many variants. See the survey [12]. Extremal results on bond-incident-degree (BID) indices are surveyed in [1], while a finite-dimensional linear framework for such indices is developed in [27].

The *Gutman–Milovanović family*, introduced in [14] and studied in the recent bounds and QSPR literature [11, 21, 25], unifies the product-based and sum-based degree indices with two parameters. It is defined by  $M_{\alpha,\beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ ,  $\alpha, \beta \in \mathbb{R}$ . Its special cases are catalogued in [21]. They include the first and second Zagreb indices  $M_{0,1} = M_1$  and  $M_{1,0} = M_2$ , the hyper-Zagreb index  $M_{0,2} = HM$  [32], the modified second Zagreb index  $M_{-1,0} = {}^mM_2$  and the Randić index  $M_{-1/2,0} = R$ . Further special cases are the sum-connectivity index  $M_{0,-1/2} = \chi$  [36], the harmonic index  $2M_{0,-1} = H$ , the inverse sum indeg index  $M_{1,-1} = ISI$ , the geometric–arithmetic index  $2M_{1/2,-1} = GA$  [34] and the arithmetic–geometric index  $\frac{1}{2}M_{-1/2,1} = AG$ . The general Randić family  $R_\alpha = M_{\alpha,0}$  and the general sum-connectivity family  $\chi_\beta = M_{0,\beta}$  are its two parameter axes, and parameter-optimised versions of the family have already been fitted to octane properties [2]. The family, its special cases and its tunability are therefore not new.

In this paper we study a three-parameter extension of  $M_{\alpha,\beta}$ , the *Loyola index family*, obtained by attaching a separately parameterised exponential factor in the normalised degree imbalance  $q_{uv} = |d_u - d_v|/(d_u + d_v)$  of each edge. The definition is  $LO(G; \alpha, \beta, \gamma) = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{\gamma q_{uv}}$ , so that  $LO(G; \alpha, \beta, 0) = M_{\alpha,\beta}(G)$  identically, and positive  $\gamma$  exponentially upweights degree-imbalanced edges while negative  $\gamma$  downweights them. On regular graphs  $\gamma$  has no effect. The paper therefore concerns  $\gamma$  alone. We ask whether the imbalance direction is independent of the parent parameters, what analytic structure it carries, and whether it adds structural or predictive information beyond an optimised Gutman–Milovanović par-

ent.

Writing the summand as  $\exp(\alpha \ln(d_u d_v) + \beta \ln(d_u + d_v) + \gamma q_{uv})$  shows that the family is a three-parameter subfamily of the exponential vertex-degree-based indices of Rada [26], whose extremal theory is developed in [10, 16, 33]. Extremal problems are not considered here. Published one-parameter exponential indices (e.g. [7, 8]) place their parameter on a single classical generator inside the exponent. Here the two Gutman–Milovanović basis terms act as the factors  $(d_u d_v)^\alpha (d_u + d_v)^\beta$ , while the imbalance coordinate carries its own parameter  $\gamma$ . The imbalance ratio itself is not new. It appears in the diminished Sombor index [22] and in the irregularity index  $IRLA(G) = 2 \sum_{uv} q_{uv}$  [30]. The pure- $\gamma$  point  $LO(G; 0, 0, \gamma) = \sum_{uv} e^{\gamma q_{uv}}$  is the exponential index of the latter in the sense of [26]. The ratio also appears in degree-ratio indices [13, 24] and in the parent family itself, since  $1 - q_{uv}^2 = 4d_u d_v / (d_u + d_v)^2$  gives  $M_{\alpha, -2\alpha}(G) = 4^{-\alpha} \sum_{uv} (1 - q_{uv}^2)^\alpha$ . What is new is a coordinate *linear* in  $q_{uv}$  in the exponent. Unlike  $\ln(1 - q_{uv}^2)$ , this coordinate is not an affine function of  $\ln(d_u d_v)$  and  $\ln(d_u + d_v)$  once a vertex of degree at least 3 is admitted. We prove this below. To our knowledge, and within the scope of the literature search documented in the reproducibility package, no published index couples it multiplicatively, with its own parameter, to the Gutman–Milovanović family. The exponential form is not derived from a physical model and is chosen instead because it makes the family log-linear in its parameters. Then  $\gamma$  acts multiplicatively and only on irregularity, and the whole family is an exponential family in  $(\alpha, \beta, \gamma)$ , which makes available the convexity, identifiability and discrimination results that follow.

On hydrogen-suppressed molecular graphs with  $\Delta \leq 4$ , every BID index is a linear functional of the ten edge-degree-pair counts  $m_{ij}(G) = |\{uv \in E(G) : \{d_u, d_v\} = \{i, j\}\}|$ ,  $1 \leq i \leq j \leq 4$  (compare the linear framework of Rada [27]). Hence the Loyola family lies inside a fixed ten-dimensional descriptor space, however its parameters are chosen.

Several of the mathematical results apply standard tools, namely log-sum-exp convexity, Laguerre’s rule of signs, the Lindemann–Weierstrass theorem and exponential-family minimality. The contribution lies in the structural characterisation they yield. The family is not proposed as a

better predictor, and we find no predictive advantage on the sets and protocols studied here. Its interest lies in the structure it exposes. It keeps the interpretation of the parent family, separates size from shape, and shares the ratio-plane geometry of  $GA$  and  $AG$ . Because the QSPR datasets are small (18 octanes, 35 nonanes, 34 decanes, 100 pooled C6–C10 alkanes), the chemometric results are diagnostic evidence rather than proof of chemical predictive utility.

The contributions are as follows. (i) The scale–shape factorisation and the ratio-plane theorem (Theorem 2). This theorem characterises exactly what the pure- $\gamma$  line can discriminate and is in large part a limitation result, with tree collisions at every order from 13, and from 16 for  $\Delta \leq 4$ . (ii) The crossing bound (Proposition 3, from Laguerre’s rule of signs) and per-graph identifiability (Proposition 4), the latter resting on an elementary independence lemma (Lemma 1). (iii) Log-convexity with mean/covariance derivatives (Theorem 1, standard), and closed forms and routine termwise bounds for  $\gamma \geq 0$  (Supplementary Material). (iv) The degree-pair ceiling, an elementary observation that is realised on the octane benchmark (Proposition 6), and structure sensitivity. (v) A negative, reproducible QSPR diagnostic, in which freeing  $\gamma$  gives no consistent gain once the parent’s search box does not truncate it. This is complemented by pre-registered tests on measured boiling points of decanes and of pooled C6–C10 alkanes (Section 5.4). There, ridge regression on the degree-pair counts is far ahead of every single index on the decanes and ahead on the pooled set.

## 2 Preliminaries

### 2.1 Notation

Let  $G = (V, E)$  be a finite simple connected graph of order  $n = |V|$  and size  $m = |E|$ , and for a vertex  $v \in V$  let the degree  $d_v = d_G(v)$  count the edges incident to  $v$ . The maximum and minimum degrees are denoted  $\Delta = \Delta(G)$  and  $\delta = \delta(G)$ . The chemometric sections use hydrogen-suppressed molecular graphs of alkane carbon skeletons, for which  $\Delta \leq 4$ , while the

mathematical results do not assume this bound unless stated.

For an edge  $e = uv \in E(G)$  we write  $(d_u, d_v)$  for its *degree pair*, understood as an unordered multiset  $\{d_u, d_v\}$ .

## 2.2 Definition

**Definition 1** (Loyola index family). Let  $G = (V, E)$  be a simple connected graph. For  $\alpha, \beta, \gamma \in \mathbb{R}$ , the *Loyola index family* is defined as,

$$LO(G; \alpha, \beta, \gamma) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta \exp\left(\gamma \frac{|d_u - d_v|}{d_u + d_v}\right) \quad (1)$$

Eq. (1) is well defined for all  $\alpha, \beta \in \mathbb{R}$  on graphs without isolated vertices. Each summand is symmetric in  $(u, v)$ , so  $LO(G; \alpha, \beta, \gamma)$  depends only on the multiset of unordered edge degree pairs of  $G$ . When  $\gamma = 0$  the family reduces identically to the Gutman–Milovanović family [14, 21],

$$LO(G; \alpha, \beta, 0) = M_{\alpha, \beta}(G) = \sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

whose classical special cases are catalogued in the Introduction. Throughout,  $q_{uv} = \text{imb}(d_u, d_v)$ , where  $\text{imb}(x, y) = |x - y|/(x + y)$ , denotes the normalised degree imbalance of the edge  $uv$ , and positive  $\gamma$  exponentially upweights degree-imbalanced edges while negative  $\gamma$  downweights them. On regular graphs every  $q_{uv} = 0$ , so  $\gamma$  has no effect and  $LO(G; \alpha, \beta, \gamma) = M_{\alpha, \beta}(G)$  for all  $\gamma$ .

## 3 Parameter geometry

Throughout,  $z_{uv} = (\ln(d_u d_v), \ln(d_u + d_v), q_{uv})$  is the feature vector of the edge  $uv$  and  $\theta = (\alpha, \beta, \gamma)$ , so that  $LO_G(\theta) = \sum_{uv \in E(G)} e^{\theta^\top z_{uv}}$ .

**Lemma 1** (Affine independence of the imbalance direction). *For  $\Delta \geq 3$ , with  $P_\Delta = \{(i, j) : 1 \leq i \leq j \leq \Delta\}$  the admissible degree pairs, the augmented feature vectors  $(1, \ln(ij), \ln(i + j), q_{ij})$ ,  $(i, j) \in P_\Delta$ , span  $\mathbb{R}^4$ . Consequently, if two parameter triples  $\theta, \theta'$  satisfy  $e^{\theta^\top z} = c e^{\theta'^\top z}$  for some*

constant  $c > 0$  and all  $(i, j) \in P_\Delta$ , then  $\theta = \theta'$  and  $c = 1$ . In particular, no choice of  $(\alpha', \beta')$  and scaling constant reproduces the action of  $\gamma \neq 0$  on all admissible degree pairs. For  $\Delta = 2$  the span has dimension 3, while for  $\Delta = 1$  the only pair is  $(1, 1)$ , with augmented vector  $(1, 0, \ln 2, 0)$ , and the span has dimension 1. In both cases the conclusion fails.

*Proof.* It suffices to show that  $q_{ij}$  is not an affine function of  $(\ln(ij), \ln(i+j))$  on  $P_\Delta$ , and the four pairs  $(1, 1), (2, 2), (1, 2), (1, 3)$ , all admissible once  $\Delta \geq 3$ , witness this. Suppose  $q_{ij} = a + b \ln(ij) + c \ln(i+j)$  on these pairs. The pairs  $(1, 1)$  and  $(2, 2)$  have  $q = 0$ , so they give  $a + c \ln 2 = 0$  and  $a + (2b + 2c) \ln 2 = 0$ , hence  $a = -c \ln 2$  and  $b = -c/2$ . Substituting into the  $(1, 2)$  and  $(1, 3)$  equations gives  $\frac{1}{3} = c(\ln 3 - \frac{3}{2} \ln 2)$  and  $\frac{1}{2} = c(\ln 2 - \frac{1}{2} \ln 3)$ . In particular  $c \neq 0$ . Three times the first minus twice the second gives  $0 = c(4 \ln 3 - \frac{13}{2} \ln 2)$ , hence  $8 \ln 3 = 13 \ln 2$ , i.e.  $3^8 = 2^{13}$ . This is false ( $6561 \neq 8192$ ). Hence the augmented vectors have rank 4. If  $e^{\theta^\top z} = c e^{\theta'^\top z}$  on  $P_\Delta$ , then  $(\theta - \theta')^\top z_{ij} = \ln c$  for all admissible pairs, so  $(\ln c, \theta' - \theta)$  is orthogonal to every augmented vector, and  $\theta = \theta'$  and  $c = 1$ . For  $\Delta = 2$  only the pairs  $(1, 1), (1, 2), (2, 2)$  exist, and a direct computation shows the augmented vectors have rank 3, while for  $\Delta = 1$  the single augmented vector has rank 1. ■

*Remark* (Independence versus per-graph identifiability). Lemma 1 concerns the edge-weight *function* on the full admissible pair set rather than every individual graph. On a single graph  $G$  some direction  $v$  satisfies  $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$  for all  $t$ , unless the realised feature vectors  $z_{ij}$  affinely span  $\mathbb{R}^3$  (Proposition 4). Along such a direction the index only rescales. On a regular graph nothing about  $\theta$  is identifiable beyond one linear functional. If the edges of  $G$  realise at most three affinely independent augmented vectors, some direction of  $\theta$ , possibly the  $\gamma$  direction, is unidentifiable from  $LO(G; \cdot)$  alone. The lemma guarantees that no such degeneracy is forced on the whole molecular class  $\Delta \geq 3$ , so the family separates parameters as a family of descriptors, although a particular graph may not.

The next result is the standard convexity of a log-sum-exp (log-partition) function, stated for  $LO$  because strictness depends on the

degree pairs that  $G$  realises.

**Theorem 1** (Global log-convexity). *Fix a graph  $G$  with  $m \geq 1$  edges, define  $F_G(\theta) = \log LO_G(\theta)$  on  $\mathbb{R}^3$ , and let  $\mathbb{E}_\theta$  and  $\text{Cov}_\theta$  denote mean and covariance under the edge probability measure  $w_{uv}(\theta) = e^{\theta^\top z_{uv}} / LO_G(\theta)$ . Then  $F_G$  is convex, with*

$$\nabla F_G(\theta) = \mathbb{E}_\theta[z], \quad \nabla^2 F_G(\theta) = \text{Cov}_\theta(z) \succeq 0.$$

*In particular  $\partial_\gamma F_G = \mathbb{E}_\theta[q] \in [0, 1)$  and  $\partial_\gamma^2 F_G = \text{Var}_\theta(q) \geq 0$ , so  $\log LO$  is nondecreasing and convex in  $\gamma$ , and it is strictly increasing in  $\gamma$  unless every edge of  $G$  joins vertices of equal degree. Moreover  $\text{Var}_\theta(q) = 0$  at some (equivalently, every)  $\theta$  if and only if  $q_{uv}$  is constant over the edges of  $G$ , which covers regular graphs ( $q \equiv 0$ ) and, for example,  $(\Delta, \delta)$ -biregular graphs. On these every edge carries the same nonzero imbalance, and  $\log LO$  is affine and strictly increasing in  $\gamma$ .  $F_G$  is strictly convex if and only if the edge feature vectors  $\{z_{uv}\}_{uv \in E}$  do not lie in a proper affine subspace of  $\mathbb{R}^3$ . On a regular graph  $F_G(\theta) = \log m + \theta^\top z_0$  is affine and  $\text{Cov}_\theta(z) = 0$ .*

The formulas follow by direct differentiation (proof in Supplementary Section S9), and the Cauchy–Schwarz midpoint inequality there is midpoint log-convexity, a corollary of this theorem.

### 3.1 The ratio plane, crossings and identifiability

Since  $4ij/(i+j)^2 = 1 - q_{ij}^2$ , every admissible pair satisfies  $\ln(ij) = 2\ln(i+j) + \ln(1 - q_{ij}^2) - \ln 4$ , and therefore

$$LO(G; \alpha, \beta, \gamma) = 4^{-\alpha} \sum_{uv \in E(G)} (d_u + d_v)^{2\alpha + \beta} (1 - q_{uv}^2)^\alpha e^{\gamma q_{uv}}. \quad (2)$$

Thus  $\kappa = 2\alpha + \beta$  acts only on the edge degree sum (scale), while the parameters  $\alpha, \gamma$  act only on  $q_{uv}$ , a function of the degree ratio (shape). Write  $n_q(G)$  for the number of edges of  $G$  with  $q_{uv} = q$  (the  $q$ -histogram) and  $\Pi = \{(\alpha, \beta, \gamma) : \beta = -2\alpha\}$  for the plane  $\kappa = 0$ , which contains every pure- $\gamma$  point  $(0, 0, \gamma)$ ,  $GA = 2LO(\frac{1}{2}, -1, 0)$  and  $AG = \frac{1}{2}LO(-\frac{1}{2}, 1, 0)$ .

**Theorem 2** (Ratio plane). *Let  $G$  and  $H$  be graphs.*

- (i) *On  $\Pi$ ,  $LO(G; \alpha, -2\alpha, \gamma) = \sum_q n_q(G) ((1 - q^2)/4)^\alpha e^{\gamma q}$  depends on  $G$  only through its  $q$ -histogram, and  $G$  and  $H$  agree at every point of  $\Pi$  if and only if  $n_q(G) = n_q(H)$  for all  $q$ . Otherwise their values differ at every point of  $\Pi$  outside a set of measure zero.*
- (ii) *For every nonzero algebraic  $\gamma$ ,  $LO(G; 0, 0, \gamma) = LO(H; 0, 0, \gamma)$  if and only if  $n_q(G) = n_q(H)$  for all  $q$ . On any finite set of graphs the same holds for all but finitely many real  $\gamma$ .*
- (iii) *Let  $G$  and  $H$  be trees with distinct edge-degree-pair profiles and equal  $q$ -histograms, each having a leaf adjacent to a vertex of the same degree  $d \geq 2$ . Attaching a pendant path of length  $L \geq 1$  at such a leaf in each produces trees with the same two properties and the same maximum degree. Consequently, for every  $n \geq 13$  (and every  $n \geq 16$  among trees with  $\Delta \leq 4$ ) there are non-isomorphic trees of order  $n$  with distinct degree-pair profiles that no point of  $\Pi$  separates.*

*Proof.* (i) is immediate from (2), since setting  $\beta = -2\alpha$  removes the degree-sum factor. For (ii) and (iii), agreement for equal histograms is immediate. Suppose the histograms differ. At  $\alpha = 0$  the difference is  $\sum_q (n_q(G) - n_q(H)) e^{\gamma q}$ , an exponential sum with distinct exponents and not all coefficients zero, so it is not identically zero. The difference on  $\Pi$  is therefore a nonzero real-analytic function of  $(\alpha, \gamma)$ , whose zero set has measure zero. (ii) For algebraic  $\gamma \neq 0$  the exponents  $\gamma q$ ,  $q \in \mathbb{Q}$ , are distinct algebraic numbers, so by the Lindemann–Weierstrass theorem [3] the numbers  $e^{\gamma q}$  are linearly independent over the algebraic numbers. Hence the integer coefficients  $n_q(G) - n_q(H)$  must all vanish. For real  $\gamma$ , a nonzero exponential sum with  $k$  distinct real exponents has at most  $k - 1$  real zeros (Proposition 3), and a finite set of graphs gives finitely many such sums. (iii) The attachment changes the profile of each tree by the same vector,  $-(1, d) + (2, d) + (L - 1)(2, 2) + (1, 2)$ , so distinct profiles stay distinct and equal histograms stay equal. The new or changed vertices have degree at most 2, so the maximum degree is unchanged when it is at least 2. The order-13 pair of Section 4.2 has a common leaf-neighbour degree  $d = 3$ , and the order-16 pair with  $\Delta \leq 4$ ,  $\{(1, 2)^2, (1, 4)^8, (2, 4)^2, (4, 4)^3\}$

and  $\{(1, 4)^8, (2, 2)^3, (2, 4)^4\}$ , has  $d = 4$ , which gives every larger order. ■

Theorem 2 is both a limitation result and a characterisation. By exhaustive enumeration of orders 7–17 (orders below 7 are trivial),  $q$ -histograms coincide with degree-pair profiles for all trees of order at most 12. The same holds for trees with  $\Delta \leq 4$  of order at most 15. So the pure- $\gamma$  points attain the degree-pair floor exactly up to those orders and, by part (iii), at no larger order. The tie of  $GA$ ,  $AG$  and the pure- $\gamma$  points on the trees of order 10 (Section 4.2) is consistent with this, since all lie on  $\Pi$ . At orders 13–17 generic points off  $\Pi$  (Proposition 7) and the Randić index separate more trees than any point of  $\Pi$ . In the other direction,  $GA$ , whose values are sums of square roots, merges two distinct  $q$ -histograms among trees first at order 17. A witness is the pair  $\{(1, 5)^{10}, (2, 2)^2, (5, 5)^2, (2, 5), (1, 2)\}$  and  $\{(4, 5)^3, (1, 4)^5, (1, 5)^6, (2, 5), (1, 2)\}$ , which have equal  $GA$  but different  $q$ -histograms, while at that order the pure- $\gamma$  points, by part (ii), still separate all histograms. That  $GA$  depends only on the degree ratio is well known, and this paper adds the exact ceiling and the colliding families.

**Proposition 3** ( $\gamma$ -crossings). *Fix  $(\alpha, \beta)$  and graphs  $G, H$ , let  $q_1 < \dots < q_r$  be the distinct imbalances of the degree pairs occurring in  $G$  or  $H$ , and write  $D(\gamma) = LO(G; \alpha, \beta, \gamma) - LO(H; \alpha, \beta, \gamma) = \sum_{k=1}^r c_k e^{\gamma q_k}$  with  $c_k = \sum_{q_{ij}=q_k} (m_{ij}(G) - m_{ij}(H))(ij)^\alpha (i+j)^\beta$ . Let  $V$  be the number of sign changes in  $(c_1, \dots, c_r)$  after zero terms are removed. If  $D \not\equiv 0$ , then  $D$  has at most  $V \leq r - 1$  real zeros counted with multiplicity, and their number has the parity of  $V$ . In particular, for graphs with  $\Delta \leq 4$  the possible imbalances are  $0, \frac{1}{7}, \frac{1}{5}, \frac{1}{3}, \frac{1}{2}, \frac{3}{5}$ , and two such graphs exchange order at most five times as  $\gamma$  ranges over  $\mathbb{R}$ , for every  $(\alpha, \beta)$ .*

*Proof.* With  $x = e^\gamma > 0$ ,  $D$  becomes the generalised polynomial  $\sum_k c_k x^{q_k}$  with distinct real exponents, so Laguerre’s extension of Descartes’ rule of signs applies [17]. The number of positive zeros, with multiplicity, is at most  $V$  and has its parity, and the parity statement also follows from the signs of  $D$  as  $\gamma \rightarrow \mp\infty$ . These are the signs of the first and last nonzero  $c_k$ , by the termwise argument of Proposition 5 applied to  $D$ . ■

The bound  $V$  can be attained. Two trees of order 12 with maximum degree 6 and eight distinct imbalances have  $V = 5$  and cross five times at  $(\alpha, \beta) = (-1, 0)$ , near  $\gamma = -17.2, -12.0, -2.0, 2.7$  and  $31.1$ . By the bound these are all its zeros. Whether five crossings are attained for  $\Delta \leq 4$  is open. Along directions other than the  $\gamma$  axis the exponents are not the  $q_k$ , so the constant five does not apply.

The per-graph identifiability discussed in the remark after Lemma 1 can be made exact. Let  $R(G)$  be the set of degree pairs realised by  $G$ ,  $r(G)$  the number of distinct imbalances among them, and  $A(G)$  the affine dimension of  $\{z_{ij} : (i, j) \in R(G)\}$ .

**Proposition 4** (Per-graph scale-invariant directions). *(i)  $A(G)$  equals the rank of the Hessian  $\text{Cov}_\theta(z)$  of  $\log LO_G$  at every  $\theta$ , and the directions  $v$  with  $LO(G; \theta + tv) = e^{ct} LO(G; \theta)$  for all  $t$  and  $\theta$  form a subspace of dimension  $3 - A(G)$ .*

*(ii)  $A(G) \leq \min(|R(G)| - 1, r(G), 3)$  for every graph.*

*(iii) If  $r(G) = 2$  with imbalances  $q_1 \neq q_2$ , then  $v = (1, -2, c)$  with  $c = [\ln(1 - q_2^2) - \ln(1 - q_1^2)] / (q_1 - q_2)$  is such a direction.*

*Proof.* (i) is the standard minimal-representation property of exponential families [6], and since  $\text{Cov}_\theta(z)$  has full support on  $R(G)$ , its rank is the affine dimension of the support. Also  $v^\top z$  is constant on  $R(G)$  exactly for  $v$  orthogonal to the differences of the  $z_{ij}$ . (ii) By (2),  $z_{ij} = (2s + \ln(1 - q^2) - \ln 4, s, q)$  with  $s = \ln(i + j)$ , so pairs of equal imbalance lie on one line parallel to  $(2, 1, 0)$ , and  $r$  parallel lines span an affine space of dimension at most  $r$ . (iii)  $v^\top z_{ij} = \ln(1 - q_{ij}^2) - \ln 4 + c q_{ij}$  takes the same value at  $q_1$  and  $q_2$  by the choice of  $c$ . ■

Equality holds in (ii) for every set of admissible pairs with  $\Delta \leq 6$  (by exhaustive enumeration), hence for all molecular graphs. It fails from  $\Delta = 7$ . The single violating set of at most four pairs there is  $\{(1, 7), (2, 6), (3, 5), (4, 4)\}$ , whose degree sums all equal 8. A constant degree sum makes  $\ln(i + j)$  constant on the set, so  $A \leq 2$ , while the bound equals 3 as soon as the set has at least four pairs and three imbalances. A connected graph all of whose edges have the same degree sum  $s$  realises

only one pair. Indeed, each neighbour of a vertex of degree  $d$  then has degree  $s - d$ , so by connectivity only the pair  $\{d, s - d\}$  occurs. Hence no connected graph has this set as its pair set. As an illustration of (iii), the spider with four legs of length three realises the pairs  $(2, 4)$ ,  $(2, 2)$  and  $(1, 2)$ , which have only two imbalances ( $\frac{1}{3}$  and 0). On it,  $\gamma$  is confounded with  $(\alpha, \beta)$  along the direction of part (iii). For a QSPR dataset the relevant set is the union of the pairs realised by its molecules. A direction along which every molecule's index is rescaled by the same factor is invisible to any correlation or regression on the dataset. This concerns identifiability of  $\theta$  from the index values on the given set of graphs rather than identifiability of a fitted regression model. On the octanes this union contains nine of the ten admissible pairs and its affine dimension is 3, so no such direction exists.

The behaviour at the two ends of the  $\gamma$  axis is used in the proof of Proposition 3. Routine termwise bounds in  $(m, \Delta, \delta)$ , a Cauchy–Schwarz interpolation bound and a uniform ratio bound are collected in Supplementary Section S9.

**Proposition 5.** *At  $\gamma = 0$ ,  $LO(G; \alpha, \beta, 0)$  coincides with the two-parameter  $(\alpha, \beta)$  degree-based index  $\sum_{uv \in E(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta$ . Let  $E_{\max}(G)$  denote the set of edges realising  $\text{imb}_{\max}(G) = \max_{uv} \text{imb}(d_u, d_v)$  and  $E_{\min}(G)$  the set realising  $\text{imb}_{\min}(G) = \min_{uv} \text{imb}(d_u, d_v)$ . Then as  $\gamma \rightarrow +\infty$ ,*

$$\frac{LO(G; \alpha, \beta, \gamma)}{e^{\gamma \text{imb}_{\max}(G)}} \longrightarrow \sum_{uv \in E_{\max}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta,$$

and as  $\gamma \rightarrow -\infty$ ,

$$LO(G; \alpha, \beta, \gamma) / e^{\gamma \text{imb}_{\min}(G)} \longrightarrow \sum_{uv \in E_{\min}(G)} (d_u d_v)^\alpha (d_u + d_v)^\beta.$$

*Proof.* At  $\gamma = 0$  the exponential factor is 1 on every edge. With  $\mu(G) := \text{imb}_{\max}(G)$  and  $\varepsilon_{uv} := \mu(G) - \text{imb}(d_u, d_v) \geq 0$ , we have  $LO(G; \alpha, \beta, \gamma) / e^{\gamma \mu(G)} = \sum_{uv} (d_u d_v)^\alpha (d_u + d_v)^\beta e^{-\gamma \varepsilon_{uv}}$ . The summands over  $E_{\max}(G)$  ( $\varepsilon_{uv} = 0$ ) do not depend on  $\gamma$ , while the others ( $\varepsilon_{uv} > 0$ ) tend to 0 as  $\gamma \rightarrow +\infty$ , termwise since  $E(G)$  is finite. The case  $\gamma \rightarrow -\infty$

## 4 Discrimination

### 4.1 The degree-pair ceiling

No vertex-degree-based index separates graphs with equal degree-pair counts, so the number of distinct profiles is a *ceiling* on distinct values and a *floor* on degeneracy.

**Proposition 6** (Degree-pair information ceiling). *If two graphs  $G$  and  $H$  have identical unordered edge-degree-pair count vectors,  $m_{ij}(G) = m_{ij}(H)$  for all  $i \leq j$ , then  $LO(G; \theta) = LO(H; \theta)$  for every  $\theta \in \mathbb{R}^3$ , and more generally  $TI(G) = TI(H)$  for every vertex-degree-based index  $TI$ . Among the 18 octane isomers exactly 16 distinct count vectors occur, and the pairs  $\{3\text{-methylheptane}, 4\text{-methylheptane}\}$  and  $\{3,4\text{-dimethylhexane}, 3\text{-ethyl-2-methylpentane}\}$  share their count vectors but differ in measured properties (Table 1). Hence every vertex-degree-based descriptor, tuned or not, carries a nonzero irreducible error on this benchmark.*

*Proof.* By Definition 1,  $LO(G; \theta)$  depends on  $G$  only through the multiset of unordered edge degree pairs, that is, through  $(m_{ij}(G))_{i \leq j}$ , and the same holds for every vertex-degree-based index [27]. The counts for the 18 octane isomers are computed and cross-checked in the reproducibility package, and Table 1 lists the common count vectors of both collision pairs. ■

The ceiling is tight. Distinct count vectors are separated at almost every parameter choice.

**Proposition 7** (Generic discrimination of distinct profiles). *Fix  $\Delta \geq 1$  and let  $m \neq m'$  be two distinct edge-degree-pair count vectors supported on the admissible pairs  $P_\Delta$ . Then  $D(\theta) = \sum_{(i,j) \in P_\Delta} (m_{ij} - m'_{ij}) e^{\theta^\top z_{ij}}$  is a real-analytic function of  $\theta$  that is not identically zero, and its zero set has Lebesgue measure zero in  $\mathbb{R}^3$ . Consequently, any two graphs with distinct count vectors are distinguished by  $LO(\cdot; \theta)$  for almost every  $\theta \in \mathbb{R}^3$ , while*

only graphs with identical count vectors (Proposition 6) are indistinguishable for all  $\theta$ .

*Proof.* The map  $(i, j) \mapsto z_{ij}$  is injective on  $P_\Delta$ , since  $\ln(ij)$  and  $\ln(i + j)$  recover the product and the sum, which determine the unordered pair. Choose  $v \in \mathbb{R}^3$  such that the inner products  $v^\top z_{ij}$  are pairwise distinct, which excludes only finitely many hyperplanes of  $v$ . Along the ray  $\theta = tv$ ,  $D(tv) = \sum_k c_k e^{t\lambda_k}$  with distinct exponents  $\lambda_k$  and coefficients  $c_k = m_{i_k j_k} - m'_{i_k j_k}$  that are not all zero, and such an exponential sum vanishes identically only if all  $c_k = 0$ . Hence  $D \not\equiv 0$  on the ray, and so on  $\mathbb{R}^3$ . Since  $D$  is a finite sum of exponentials of linear functionals, it is real-analytic on  $\mathbb{R}^3$ , and the zero set of a nonzero real-analytic function has Lebesgue measure zero.  $\blacksquare$

The proof uses only the injectivity of  $(i, j) \mapsto (\ln(ij), \ln(i + j))$  and therefore applies to the parent  $M_{\alpha, \beta}$  on the  $(\alpha, \beta)$ -plane, so Proposition 7 is inherited from the parent rather than conferred by  $\gamma$ . Since  $q_{ij} = \sqrt{1 - 4ij/(i + j)^2}$  depends only on the pair,  $\gamma$  adds no graph information beyond the edge-degree-pair counts, of which every vertex-degree-based index is a functional [27]. What it adds is the functional direction of Lemma 1, that is, a different weighting of the same information. As a corollary, on graphs of maximum degree four any ten degree-based indices with linearly independent weight vectors  $(f(i, j))_{i \leq j \leq 4}$  span the ten-dimensional count space. Hence every bond-incident-degree index, the Loyola family at every parameter value included, is an exact linear combination of them and adds no linear information for any target. Whether such a reweighting still helps prediction is the question of Section 5.

## 4.2 Degeneracy across tree orders

Degeneracy [18] measures the failure to discriminate non-isomorphic graphs, and we quantify it with the distinct-value convention  $100(1 - k/N)$ , where  $k$  is the number of distinct values among  $N = 106$  trees (Supplementary Section S5). On the 106 trees of order 10 the pure- $\gamma$  points  $LO(0, 0, 1)$  and  $LO(0, 0, 2)$  attain 25.5% (79 distinct values), tied with  $R$ ,  $GA$  and  $AG$ . The 106 trees realise only 79 distinct edge-degree-pair

**Table 1.** The two degree-pair collision pairs among the 18 octane isomers. Each pair shares its complete edge-degree-pair count vector, shown as multiplicities of the unordered pairs, and is therefore indistinguishable by every vertex-degree-based index for all parameter values. The two members differ in the measured properties.

| Isomer                                                         | $T_B$ | $\Delta H_f$ | $\Delta H_{\text{vap}}$ | $S$    | $\omega$ |
|----------------------------------------------------------------|-------|--------------|-------------------------|--------|----------|
| <i>shared count vector</i> 2(1, 2), 1(1, 3), 2(2, 2), 2(2, 3): |       |              |                         |        |          |
| 3-methylheptane                                                | 118.9 | -50.82       | 9.52                    | 111.26 | 0.371    |
| 4-methylheptane                                                | 117.7 | -50.69       | 9.48                    | 109.32 | 0.372    |
| <i>shared count vector</i> 2(1, 2), 2(1, 3), 2(2, 3), 1(3, 3): |       |              |                         |        |          |
| 3,4-dimethylhexane                                             | 117.7 | -50.91       | 9.32                    | 106.59 | 0.340    |
| 3-ethyl-2-methylpentane                                        | 115.6 | -50.48       | 9.21                    | 106.06 | 0.330    |

count vectors, so 25.5% is the floor for *any* degree-pair-based index, while  $M_1$  (83.0%) and  $M_2$  (67.0%) are the most degenerate (Supplementary Figure S2). For orders 7 through 12, on all trees and on those with  $\Delta \leq 4$  (Supplementary Table S4), the pure- $\gamma$  points and  $R$  attain the floor at every order. The index  $\chi$  leaves it from order 10, with 77 values against 79 profiles. Theorem 2 makes this exact. The pure- $\gamma$  points separate precisely the  $q$ -histograms, which coincide with the degree-pair profiles up to order 12 (order 15 for  $\Delta \leq 4$ ) and not beyond. At order 13 the profiles  $\{(1, 2)^1, (1, 3)^2, (1, 6)^5, (2, 3)^2, (2, 6)^1, (3, 3)^1\}$  and  $\{(1, 3)^3, (1, 6)^5, (2, 2)^1, (2, 3)^2, (3, 6)^1\}$  (exponents are edge multiplicities) share the  $q$ -histogram  $\{0^1, \frac{1}{5}^2, \frac{1}{3}^1, \frac{1}{2}^3, \frac{5}{7}^5\}$ . Table 2 collects the first failures, and the index  $R$  still attains the floor at order 13 (570 values) and leaves it at order 14, where the pure- $\gamma$  points give 1090 values.

Off the ratio plane the imbalance factor can restore discrimination lost at particular classical exponents. A point  $(0, \beta, 0)$  weights an edge by its degree sum only, so pairs with equal sum receive equal weight. With  $\gamma \neq 0$  their imbalances separate them. On all trees of order 16 (4069 profiles),  $LO(0, \frac{1}{2}, 0)$  and  $LO(0, \frac{1}{3}, 0)$  take only 3475 and 3476 distinct values, while  $LO(0, \frac{1}{2}, 1)$  and  $LO(0, \frac{1}{3}, 1)$  separate all 4069.  $LO(\frac{1}{2}, 0, \cdot)$  rises from 3940 to 4067 and the Randić point  $LO(-\frac{1}{2}, 0, \cdot)$  from 4007 to 4068 as  $\gamma$  goes from 0 to 1. Points on  $\Pi$  such as  $GA$  are unaffected (3996 in both cases), as Theorem 2 requires, and generic parent points such as  $LO(0.3, 0.2, 0)$

**Table 2.** First tree order at which discrimination is lost. Values are counted as distinct at 11 significant digits. “Below floor” is the first order with fewer distinct values than distinct edge-degree-pair profiles, the floor for every vertex-degree-based index, and the entry gives (values/profiles) at that order. “Merges” is the first order at which an index on  $\Pi$  gives equal values to two trees with different  $q$ -histograms. This does not apply to  $R$ , which is not on  $\Pi$ .  $LO(0, 0, \gamma)$  denotes the pure- $\gamma$  points  $\gamma = 1, 2$ . Orders below 7 were not examined. A dash entry means not determined.

| Index              | all trees                 |                    | trees with $\Delta \leq 4$ |                    |
|--------------------|---------------------------|--------------------|----------------------------|--------------------|
|                    | below floor               | merges             | below floor                | merges             |
| $LO(0, 0, \gamma)$ | 13 (566/570)              | never <sup>a</sup> | 16 (809/810)               | never <sup>a</sup> |
| $R$                | 14 (1095/1100)            | n/a <sup>b</sup>   | > 16 <sup>c</sup>          | n/a <sup>b</sup>   |
| $GA$               | 13 (566/570) <sup>d</sup> | 17 <sup>e</sup>    | 16 (809/810) <sup>d</sup>  | — <sup>e</sup>     |

<sup>a</sup> All  $q$ -histograms are separated for every nonzero algebraic  $\gamma$  (Theorem 2(ii)). <sup>b</sup>  $R$  is not on  $\Pi$ . <sup>c</sup>  $R$  attains the floor for trees with  $\Delta \leq 4$  at every order up to 16, the largest examined. <sup>d</sup>  $GA$  lies on  $\Pi$  and merges no  $q$ -histograms below order 17, so there it ties the pure- $\gamma$  points. <sup>e</sup> The witness (Section 3.1) has maximum degree 5.

already separate all 4069 profiles (Proposition 7). The effect is established by enumeration for orders 12–16 but is not proved. It gives no predictive gain over a tuned parent in Section 5. At fixed fractional exponents the sign of the change depends on the property (Supplementary Section S5).

### 4.3 Structure sensitivity

The structure-sensitivity criterion of Furtula, Gutman and Dehmer [9], with a fingerprint-based variant due to Rakić and Furtula [28], was implemented with the published protocol on the 75 decane isomer trees. For a tree, the similar structures are those obtained by relocating a single edge within the molecular class.  $SS$  and  $Abr$  are the class means of the mean and the maximum relative change of the index over these neighbours. The implementation reproduces the published decane-class values of Barman and Das [4], with eleven of fourteen  $SS/Abr$  values agreeing exactly at the four displayed decimals. Each of the other three agrees within one unit in the fourth. The protocol and this control are detailed in Supplementary Section S6. Under the published desideratum ( $SS$  large,  $Abr$  small),  $LO(0, 0, 2)$  has a structure sensitivity indistinguishable at three

decimals from that of the hyper-Zagreb index (0.1257 against 0.1256). Its abruptness is markedly smaller (0.274 against 0.313). The sensitivity-to-abruptness ratios of the pure- $\gamma$  points (0.4579 and 0.4565) are comparable to those of  $H/2$ ,  $GA$  and  $R$  (0.448–0.451, Table 3). In view of the redundancy noted in Section 5.1, these third-decimal differences do not rank the indices.  $SS$  is also not a fixed property of the family. Along the pure- $\gamma$  line it grows with  $\gamma$  (0.031, 0.062, 0.126, 0.319 at  $\gamma = 0.5, 1, 2, 5$ ), while the ratio peaks near  $\gamma = 2$  and then falls (0.427 at  $\gamma = 5$ ). The pure- $\gamma$  points reduce to first order in  $\gamma$  to the parameter-free aggregate  $IRLA$ , since  $LO(0, 0, \gamma) = m + \frac{\gamma}{2} IRLA + O(\gamma^2)$ , and this aggregate has a higher ratio ( $SS = 0.2343$ ,  $Abr = 0.5066$ , ratio 0.4625). Hence the favourable sensitivity–abruptness balance is a property of linear imbalance weighting rather than of the exponential form. The equality of  $SS$  with hyper-Zagreb at  $\gamma = 2$  reflects the chosen value of  $\gamma$ .

**Table 3.** Structure sensitivity  $SS$ , abruptness  $Abr$ , and their ratio on the 75 decane isomers under the published similar-structure protocol of [9] (single-edge relocation within the molecular class).

| Index         | $SS$   | $Abr$  | $SS/Abr$ |
|---------------|--------|--------|----------|
| $M_1$         | 0.0566 | 0.1336 | 0.4238   |
| $M_2$         | 0.0922 | 0.2443 | 0.3776   |
| $HM$          | 0.1256 | 0.3126 | 0.4017   |
| ${}^mM_2$     | 0.0516 | 0.1177 | 0.4384   |
| $R$           | 0.0264 | 0.0588 | 0.4481   |
| $\chi$        | 0.0264 | 0.0591 | 0.4463   |
| $H/2$         | 0.0516 | 0.1143 | 0.4514   |
| $ISI$         | 0.0226 | 0.0597 | 0.3795   |
| $GA$          | 0.0241 | 0.0535 | 0.4507   |
| $AG$          | 0.0256 | 0.0579 | 0.4415   |
| $LO(0, 0, 1)$ | 0.0624 | 0.1367 | 0.4565   |
| $LO(0, 0, 2)$ | 0.1257 | 0.2745 | 0.4579   |

## 5 QSPR diagnostics

In Tables 6–8, medians are taken over candidate-stream seeds, and seed counts summarise the variability of the random search on fixed data and

are not replications. Values are computed in double precision and rounded once for display (three decimals for correlations,  $Q^2$  and RMSE, four for structure sensitivity). A tabulated difference may therefore differ in the last place from the difference of two displayed entries, and a quoted median of differences need not equal the difference of medians. The narrow- and wide-box streams use the same uniform variates scaled by the half-width, so comparisons across boxes are paired.

We benchmark the Loyola family against the classical indices it reduces to on the 18 octane isomers, following the chemometric benchmark of the diminished Sombor study [22]. The 35 nonane isomers are a replication set, with models refitted on it. The comparison set consists of the exact reductions of Section 1,  $M_1, M_2, HM, {}^mM_2, R, \chi, H/2, ISI, GA, AG$ , and the two fixed pure- $\gamma$  benchmark points  $LO(0, 0, 1)$  and  $LO(0, 0, 2)$ . Since the edge count  $m = LO(0, 0, 0)$  is constant on the octanes, it is omitted. No parameter in the correlation comparison is tuned against the property data. An exploratory local-tuning check (Supplementary Section S2) did not exceed the best fixed benchmark except by a 0.001 in-sample  $\Delta H_{\text{vap}}$  margin. All comparisons are descriptive. On sets this small they indicate how imbalance information changes the fit of the parent family, and they do not establish chemical predictive utility.

## 5.1 Correlations on the octane isomers

Table 5 reports the Pearson correlation  $r$  between each index and each of the five physicochemical properties of the 18 octane isomers. The properties are the boiling point  $T_B$ , the enthalpy of formation  $\Delta H_f$ , the enthalpy of vaporization  $\Delta H_{\text{vap}}$ , the entropy  $S$  and the acentric factor  $\omega$ . Data and units are in Table 4. Boiling points, formation enthalpies and (with one exception) vaporization enthalpies are traced to the NIST WebBook [20], and acentric factors to KDB-derived compilations. The WebBook lists a gas-phase entropy for only five isomers, while seven entropies are compared with the 1947 API Research Project 44 tables [31] or a compiled octane dataset. All 90 molecule-level values were audited against authoritative sources (Supplementary Section S1). Three caveats remain. The 298 K vaporization enthalpy of 2,2,3,3-tetramethylbutane, a solid at 298 K

(triple point 374 K), is the benchmark-compilation value  $8.41 \text{ kcal mol}^{-1}$  ( $35.19 \text{ kJ mol}^{-1}$ ). This matches neither of the NIST standard vaporization enthalpies ( $42.94$  and  $42.91 \text{ kJ mol}^{-1}$ , about  $10.26 \text{ kcal mol}^{-1}$ ), so its physical reference state is unresolved. Six gas-phase entropies could not be verified against a primary source, and the 1947 API Project 44 values differ from them by up to 2.2 units. The 2,3,3-trimethylpentane value is a compiled-dataset value, and four entropies differ from a widely used compilation. Substituting the compilation values changes all twelve  $S$  correlations (by up to 0.043 in  $|r|$ ) and moves the margin between  $HM$  and  $M_1$  at the top from 0.001 to 0.007. Margins of this size are not meaningful at this resolution. Alternative compiled acentric factors change several  $\omega$  correlations in the third decimal (e.g.  $M_2$  from  $-0.988$  to  $-0.986$ ) and leave the  $\omega$  ranking unchanged (detail in Supplementary Section S10).

Table 4 lists the full 18-isomer dataset used in this section, with  $T_B$  in  $^{\circ}\text{C}$ ,  $\Delta H_f$  and  $\Delta H_{\text{vap}}$  in  $\text{kcal mol}^{-1}$  ( $\Delta H_{\text{vap}}$  at 298 K),  $S$  in  $\text{cal mol}^{-1}\text{K}^{-1}$  and  $\omega$  dimensionless. The  $\Delta H_{\text{vap}}$  values of 3,3-dimethylhexane (8.97), 3-ethyl-3-methylpentane (9.08), 2,2,3-trimethylpentane (8.83) and 2,3,3-trimethylpentane (8.90) are values consistent with the NIST WebBook determinations, and Supplementary Section S1 gives the exact-versus-within-uncertainty status of each. The  $\Delta H_{\text{vap}}$  entry of 2,2,3,3-tetramethylbutane is the benchmark-compilation value discussed above. The  $S$  value of 2,3,3-trimethylpentane (102.06) is a compiled-dataset value (Supplementary Section S10).

Table 4 has several open provenance items, namely the unverified entropies, the vaporization enthalpy of 2,2,3,3-tetramethylbutane, the corrected transcription of the 2,3,3-trimethylpentane entropy and the alternative compiled values. These items, with their effect on Table 5, are collected in Supplementary Section S10, and none of them enters the pre-registered tests of Sections 5.3 and 5.4.

**Table 4.** Physicochemical properties of the 18 octane isomers.

| Isomer                    | $T_B$ | $\Delta H_f$ | $\Delta H_{\text{vap}}$ | $S$    | $\omega$ |
|---------------------------|-------|--------------|-------------------------|--------|----------|
| <i>n</i> -octane          | 125.6 | -49.82       | 9.92                    | 111.55 | 0.398    |
| 2-methylheptane           | 117.6 | -51.50       | 9.48                    | 109.84 | 0.378    |
| 3-methylheptane           | 118.9 | -50.82       | 9.52                    | 111.26 | 0.371    |
| 4-methylheptane           | 117.7 | -50.69       | 9.48                    | 109.32 | 0.372    |
| 2,2-dimethylhexane        | 106.8 | -53.71       | 8.92                    | 103.13 | 0.339    |
| 2,3-dimethylhexane        | 115.6 | -51.13       | 9.27                    | 108.02 | 0.348    |
| 2,4-dimethylhexane        | 109.4 | -52.44       | 9.03                    | 106.98 | 0.344    |
| 2,5-dimethylhexane        | 109.1 | -53.21       | 9.05                    | 105.72 | 0.357    |
| 3,3-dimethylhexane        | 111.9 | -52.61       | 8.97                    | 104.74 | 0.322    |
| 3,4-dimethylhexane        | 117.7 | -50.91       | 9.32                    | 106.59 | 0.340    |
| 3-ethyl-2-methylpentane   | 115.6 | -50.48       | 9.21                    | 106.06 | 0.330    |
| 3-ethyl-3-methylpentane   | 118.3 | -51.38       | 9.08                    | 101.48 | 0.302    |
| 3-ethylhexane             | 118.5 | -50.40       | 9.48                    | 109.43 | 0.362    |
| 2,2,3-trimethylpentane    | 109.8 | -52.61       | 8.83                    | 101.31 | 0.300    |
| 2,2,4-trimethylpentane    | 99.2  | -53.57       | 8.40                    | 101.81 | 0.305    |
| 2,3,3-trimethylpentane    | 114.8 | -51.73       | 8.90                    | 102.06 | 0.291    |
| 2,3,4-trimethylpentane    | 113.5 | -51.97       | 9.01                    | 102.39 | 0.317    |
| 2,2,3,3-tetramethylbutane | 106.5 | -53.99       | 8.41                    | 93.06  | 0.247    |

**Table 5.** Pearson correlation  $r$  between the Loyola reductions, the fixed pure- $\gamma$  benchmark points and the five physicochemical properties of the 18 octane isomers. Bold flags the largest  $|r|$  per column, and this is descriptive. Margins to the runner-up are 0.001–0.032, and their paired bootstrap intervals include zero (see the text).

| Index                             | $T_B$         | $\Delta H_f$  | $\Delta H_{\text{vap}}$ | $S$           | $\omega$      |
|-----------------------------------|---------------|---------------|-------------------------|---------------|---------------|
| $M_1 = LO(0, 1, 0)$               | -0.718        | -0.762        | -0.935                  | -0.970        | -0.970        |
| $M_2 = LO(1, 0, 0)$               | -0.497        | -0.542        | -0.811                  | -0.934        | <b>-0.988</b> |
| $HM = LO(0, 2, 0)$                | -0.654        | -0.710        | -0.903                  | <b>-0.971</b> | -0.982        |
| ${}^mM_2 = LO(-1, 0, 0)$          | <b>+0.855</b> | <b>+0.891</b> | +0.925                  | +0.863        | +0.794        |
| $R = LO(-1/2, 0, 0)$              | +0.819        | +0.850        | +0.958                  | +0.932        | +0.898        |
| $\chi = LO(0, -1/2, 0)$           | +0.800        | +0.832        | +0.961                  | +0.947        | +0.924        |
| $H/2 = LO(0, -1, 0)$              | +0.821        | +0.849        | +0.961                  | +0.932        | +0.900        |
| $ISI = LO(1, -1, 0)$              | -0.005        | -0.000        | -0.383                  | -0.593        | -0.738        |
| $GA = 2 LO(1/2, -1, 0)$           | +0.822        | +0.858        | +0.965                  | +0.939        | +0.906        |
| $AG = \frac{1}{2} LO(-1/2, 1, 0)$ | -0.816        | -0.859        | -0.957                  | -0.938        | -0.900        |
| $LO(0, 0, 1)$                     | -0.828        | -0.832        | <b>-0.984</b>           | -0.928        | -0.929        |
| $LO(0, 0, 2)$                     | -0.829        | -0.846        | -0.981                  | -0.937        | -0.925        |

The best correlation per property is split, with  ${}^mM_2 = LO(-1, 0, 0)$

strongest on  $T_B$  and  $\Delta H_f$ ,  $HM = LO(0, 2, 0)$  on  $S$  and  $M_2 = LO(1, 0, 0)$  on  $\omega$ . The fixed pure- $\gamma$  point  $LO(0, 0, 1)$  is strongest on  $\Delta H_{\text{vap}}$  ( $|r| = 0.984$ , with  $LO(0, 0, 2)$  second at 0.981), and at the exact-reduction triples the Loyola correlations coincide with the classical indices, as they must. For each property a descriptive post-selection paired bootstrap interval for the observed top-two  $|r|$  difference included zero, but these intervals are not multiplicity-adjusted and do not establish equivalence. Recomputing each bolded correlation with single isomers deleted shifts  $|r|$  by at most 0.027 (bootstrap details are in Supplementary Section S1). The bolded entries are the largest of twelve highly correlated sample correlations on 18 molecules, and are therefore optimistic (winner’s-curse) estimates of the selected descriptor’s correlation on new molecules. They are descriptive rankings rather than tests.

Fourteen of the descriptor pairs have  $|r| \geq 0.99$ , and the first two principal components carry 99.05% of the variance (Supplementary Section S1). The pure- $\gamma$  points are collinear with the classical reductions (maximum  $|r|$  with a reduction equal to 0.982 for  $LO(0, 0, 1)$  and 0.993 for  $LO(0, 0, 2)$ ). Differences in the third decimal of  $|r|$  between near-collinear descriptors are therefore not meaningful rankings.

Proposition 6 is realised on these data, where the 18 isomers realise only 16 distinct edge-degree-pair count vectors. The two collision pairs of Table 1 are indistinguishable by *every* vertex-degree-based index at *every* parameter choice. Their measured properties differ, by up to 1.2 °C in  $T_B$  within the first pair and 2.1 °C within the second. Every correlation in Table 5, and every tuned variant below, operates above this irreducible floor.

## 5.2 Effect of freeing $\gamma$ and of the search box

We ask whether, at identical search budgets, freeing  $\gamma$  improves out-of-sample prediction beyond an optimised parent  $M_{\alpha, \beta}$ . The design is a fully nested leave-one-out (details in Supplementary Section S3) in which, for each property and each outer fold, candidate parameter vectors are drawn uniformly per coordinate from a search box. These are triples  $(\alpha, \beta, \gamma)$  for the Loyola model and pairs  $(\alpha, \beta)$  with  $\gamma = 0$  for the parent, with

identical budget. Candidates are scored on the training isomers only, by *inner* leave-one-out root-mean-square error of the one-descriptor linear regression, after which the best candidate is refit and the held-out isomer predicted. No quantity computed on the held-out isomer enters selection or fitting. In the box  $(-2, 2)$  used first, freeing  $\gamma$  appeared to help the vaporization enthalpy (Supplementary Table S3), improving nested predictive performance for  $\Delta H_{\text{vap}}$  in every seed over one hundred candidate streams. The count was 100/100 at budgets 200 and 500, with median  $\Delta Q^2 = +0.069$  and  $+0.068$ , whereas freeing  $\gamma$  usually degraded  $\Delta H_f$  and gave mixed or slightly negative results for the other properties. With 2,2,3,3-tetramethylbutane excluded the count falls to 93/100 and 96/100 (Supplementary Section S3), and these counts describe optimiser variance on one fixed set of 18 molecules. Selection also varies across folds. For  $\Delta H_f$  the three-parameter search chose three distinct triples over the 18 folds, whereas for  $\Delta H_{\text{vap}}$  it chose the same triple in every fold (Supplementary Section S3). The nested  $Q^2$  therefore evaluates the selection-plus-fit procedure rather than one fitted model. These counts also hide a boundary effect, since for  $\Delta H_{\text{vap}}$  the parent chose  $(\alpha, \beta) = (1.021, -1.993)$ , on the boundary of the box, in 16 of the 18 folds.

The decisive design choice is the box  $(-2, 2)$ , which the nonane pre-registration also fixed. By  $\ln(ij) = 2\ln(i+j) + \ln(1-q^2) - \ln 4$  (Eq. (2)) the parent family reaches imbalance information through  $(1-q^2)^\alpha$  along the direction  $\beta \approx -2\alpha$ . It needs large exponents to weight it strongly, and the box truncates exactly that direction. The matched comparison was therefore repeated with 2000 candidates per model, seeds 0–49, and boxes of half-width 2 and 12 for both models (Table 6), a repetition that was not pre-registered. In the box of half-width 12 the parent’s selections are interior in more than 90% of folds for every property, and the vaporization-enthalpy gain shrinks to about zero. A hybrid control that widens only  $(\alpha, \beta)$  gives median  $\Delta Q^2 = +0.004$  on the octanes (28 of 50 seeds) and  $-0.002$  on the nonanes (24 of 50). Widening all three coordinates gives  $-0.010$  and  $-0.016$ , while on the other properties no advantage is consistent across boxes. Under the hybrid control the Loyola model is ahead for octane  $S$  (38 of 50 seeds) and nonane  $T_B$  (31 of 50), and behind

for octane  $T_B$ ,  $\Delta H_f$  and  $\omega$ . The wide box itself lowers the parent’s  $Q^2$  for  $S$  and  $T_B$ . The vaporization-enthalpy gains above are thus a truncation effect rather than information the parent lacks. Most of the change comes from the parent improving in the wide box ( $0.887 \rightarrow 0.942$  on the octanes). The rest comes from the three-dimensional search being slightly less efficient there ( $0.957 \rightarrow 0.946$  for the hybrid control). What remains is due to linear imbalance weighting itself. Both the fixed point  $LO(0,0,1)$  and the parameter-free *IRLA* approach what the parent reaches only near  $(\alpha, \beta) \approx (4.5, -9)$  ( $Q^2 = 0.957$  and  $0.948$  against  $0.949$  there, on the octanes).

**Table 6.** Sensitivity of the matched comparison to the search box. Oct.: octanes,  $n = 18$ . Non.: nonanes. Edge: share of outer folds in which the parent’s selection lies within 10% of the boundary of the box of the stated half-width. Medians of  $Q^2$  are over seeds 0–49 at 2000 candidates per model. Counts: the number of seeds (of 50) in which LO beats GM (half-width 2) or  $LO_h$  beats GM (half-width 12).  $LO_h$  is a hybrid control with  $(\alpha, \beta)$  in the wide box and  $\gamma$  in  $(-2, 2)$ , which avoids diluting the Loyola search over a wide  $\gamma$  range.

| Set, prop.                   | half-width 2 |       |       |    | half-width 12 |       |       |        |    |
|------------------------------|--------------|-------|-------|----|---------------|-------|-------|--------|----|
|                              | Edge         | GM    | LO    | >  | Edge          | GM    | LO    | $LO_h$ | >  |
| oct. $T_B$                   | 69%          | 0.665 | 0.730 | 36 | 2%            | 0.606 | 0.495 | 0.544  | 21 |
| oct. $\Delta H_f$            | 52%          | 0.821 | 0.769 | 11 | 0%            | 0.820 | 0.670 | 0.761  | 10 |
| oct. $\Delta H_{\text{vap}}$ | 97%          | 0.887 | 0.957 | 50 | 3%            | 0.942 | 0.934 | 0.946  | 28 |
| oct. $S$                     | 85%          | 0.904 | 0.891 | 11 | 4%            | 0.842 | 0.838 | 0.867  | 38 |
| oct. $\omega$                | 0%           | 0.979 | 0.974 | 10 | 0%            | 0.979 | 0.938 | 0.951  | 7  |
| non. $T_B$                   | 61%          | 0.576 | 0.601 | 30 | 0%            | 0.524 | 0.392 | 0.577  | 31 |
| non. $\Delta H_{\text{vap}}$ | 93%          | 0.855 | 0.895 | 49 | 9%            | 0.892 | 0.881 | 0.889  | 24 |

Under these protocols, then, tuning  $\gamma$  yields no demonstrated predictive advantage over a parent evaluated over the wider search box. Equal candidate budgets do not guarantee equally thorough search in two and three dimensions, and the hybrid control, which keeps  $\gamma$  in a narrow range, reduces this asymmetry without removing it. A more exhaustive three-dimensional search could in principle find a better Loyola point. The value of  $\gamma$  lies in the structural results of Section 3 and in the comparison with linear imbalance weighting of Section 5.3.

### 5.3 Replication on the nonane isomers

Under a protocol committed to the reproducibility package before any nonane value was collected, the identical nested design was applied to the 35 constitutional nonane isomers. Boiling points were 35 values: 20 NIST averages, 5 single determinations and 10 most recent determinations selected under the protocol’s Deviation D1. Standard vaporization enthalpies were 34 values from the NIST WebBook [20]. Only 4 of the 34 vaporization enthalpies are calorimetric determinations (one more is a NIST average), and 29 come from two compilations [19, 35]. Agreement with them therefore measures consistency with compiled values rather than independent experimental validation. Within-fold baselines are in Table 7. The pre-registered verdict is that the  $\gamma$  effect *replicates* for  $\Delta H_{\text{vap}}$  (compilation values), with LO better than GM in 95/100 and 98/100 seeds and median  $\Delta Q^2 = +0.048$  and  $+0.053$ . For  $T_B$  the verdict is *mixed* (58/100 and 63/100). The pre-registered protocol, however, fixed the box of half-width 2, which binds the parent (Table 6), and in a box that does not, the nonane gain disappears (median  $\Delta Q^2 = -0.002$  under the hybrid control). The pre-registered verdict is reported as obtained, with this deviation recorded. On both sets the untuned fixed point  $LO(0,0,1)$  matches or beats the tuned Loyola model (0.957 and 0.903 against medians 0.953 and 0.900). It is closely related to *IRLA*:  $LO(G;0,0,\gamma) = m + \frac{\gamma}{2} IRLA(G) + O(\gamma^2)$  as  $\gamma \rightarrow 0$ . At  $\gamma = 1$  higher moments of the imbalance remain, and trees with equal *IRLA* can differ in  $LO(0,0,1)$ . Even so, the correlation between the two is 0.996 on both sets. *IRLA* itself comes within 0.003 of the median Loyola model on the nonanes (0.897 against 0.900). What matters for this property is therefore a term linear in the imbalance. Ridge regression on the full degree-pair count vector predicts the nonanes better than every index model ( $Q^2 = 0.927$  for  $\Delta H_{\text{vap}}$ , 0.682 for  $T_B$ ). This is consistent with Proposition 6 and the remark after Proposition 7 (every index in the family is a one-dimensional reweighting of the degree-pair counts). A post hoc molecule bootstrap, conditional on the fitted predictions, omits re-fitting and selection uncertainty, and its 95% interval for LO minus GM on nonane  $\Delta H_{\text{vap}}$ ,  $[-0.006, +0.113]$  at budget 200, includes zero. Pro-

tolcol details, an exploratory transfer analysis and this bootstrap are in Supplementary Section S4.

**Table 7.** Nested leave-one-out  $Q^2$  on the octanes (oct.,  $n = 18$ ) and the nonane replication set (non.,  $n = 35$  for  $T_B$ , 34 for  $\Delta H_{\text{vap}}$ ). Mean: training-fold mean. Class.: best of the ten classical indices, selected inside each fold. IRLA: one-descriptor model on IRLA. Ridge: ridge regression on the edge-degree-pair counts. LO<sub>1</sub>: the fixed, untuned point  $LO(0, 0, 1)$ . GM, LO: median of  $Q^2$  over seeds 0–99 at budget 200 in the box of half-width 2 (see Table 6 for wider boxes). Last column: seeds in which LO beat GM at budgets 200/500.

| Set, property                | Mean   | Class. | IRLA  | Ridge | LO <sub>1</sub> | GM    | LO    | LO>GM   |
|------------------------------|--------|--------|-------|-------|-----------------|-------|-------|---------|
| oct. $T_B$                   | −0.121 | 0.593  | 0.585 | 0.344 | 0.591           | 0.662 | 0.652 | 53/64   |
| oct. $\Delta H_f$            | −0.121 | 0.746  | 0.593 | 0.654 | 0.639           | 0.831 | 0.689 | 14/24   |
| oct. $\Delta H_{\text{vap}}$ | −0.121 | 0.884  | 0.948 | 0.954 | 0.957           | 0.888 | 0.953 | 100/100 |
| oct. $S$                     | −0.121 | 0.905  | 0.733 | 0.721 | 0.785           | 0.904 | 0.889 | 34/31   |
| oct. $\omega$                | −0.121 | 0.971  | 0.810 | 0.851 | 0.823           | 0.979 | 0.970 | 26/28   |
| non. $T_B$                   | −0.060 | 0.502  | 0.400 | 0.682 | 0.409           | 0.577 | 0.586 | 58/63   |
| non. $\Delta H_{\text{vap}}$ | −0.062 | 0.857  | 0.897 | 0.927 | 0.903           | 0.853 | 0.900 | 95/98   |

## 5.4 Measured boiling points

Measured vaporization enthalpies are too scarce for a larger test. NIST lists only 25 acyclic alkane skeletons outside  $C_8$  and  $C_9$ , and one of the 75 decanes, with at least one calorimetric standard-state 298 K determination (NIST averaged entries resolved against their individual measurements). Two further tests therefore used measured boiling points, under a protocol committed to the repository before any model was fitted. The NIST pages had been archived two days earlier during a data-availability audit that counted available values only. Boiling points were taken from the NIST WebBook [20] by a fixed rule. The rule takes the NIST average, or else the median of non-compilation determinations if they agree within 3 K, and otherwise excludes the molecule. This gives all hexanes, heptanes and octanes and 34 of the 35 nonanes, since one nonane has two NIST entries with the same InChI whose boiling points differ by 6.4 K. It also gives 34 of the 75 decanes, because 31 of them have no NIST boiling point and 10 have determinations more than 3 K apart. For 14 of the

34 included decanes NIST lists a single non-compilation determination, so the agreement rule does not apply to them. Under this uniform rule the pooled octane boiling points can differ from those of Table 4 by up to  $0.5^{\circ}\text{C}$ , for example 2,4-dimethylhexane at  $108.9$  against  $109.4^{\circ}\text{C}$ , a documented source variation. Eight pooled nonane boiling points differ from the replication-set values of Section 5.3 by  $0.1\text{--}0.9^{\circ}\text{C}$ , as that protocol took the most recent determination where NIST gives no average. The included and excluded decanes do not differ detectably in branching or imbalance. The measures were the numbers of tertiary and quaternary carbons and of methyl groups, the Wiener index, *IRLA*, *LO*(0, 0, 1) and *R*. Two-sided permutation tests gave  $p \geq 0.10$  in every case, though with 34 and 41 molecules such tests can rule out only large differences. Test A evaluates the models on the decanes as a new cohort, refitting and reselecting inside each leave-one-out fold. Test B pools all 100 molecules and adds the carbon number  $n_C$  as a covariate, so that each descriptor is judged on branching beyond size. Both use the non-truncating design of Section 5.2, with the parent searched over  $(\alpha, \beta) \in (-12, 12)^2$  and the Loyola model over the same box with  $\gamma \in (-2, 2)$ , at 2000 candidates and seeds 0–49. The pre-registered criterion for a  $\gamma$  effect required at least 40 wins in 50 seeds and a molecule-bootstrap interval excluding zero (Table 8). The rule is one-sided, so a negative difference also returns “no evidence” and the rule never returns evidence that  $\gamma$  hurts.

**Table 8.** Measured boiling points (nested leave-one-out  $Q^2$ ). Test A: decane cohort, refitted within folds. Test B: pooled C6–C10 with the carbon number  $n_C$  in every model (Class., *IRLA*, *LO*<sub>1</sub>:  $n_C$  plus that descriptor). Base: training-fold mean (Test A) or  $n_C$  alone (Test B). Class.: best classical index chosen in each fold. *LO*<sub>1</sub>: fixed *LO*(0, 0, 1). Ridge: ridge regression on the degree-pair counts. GM, *LO*: medians over seeds 0–49 in the non-truncating boxes.  $\Delta Q^2$ : median of *LO* minus GM. Last column: seeds (of 50) in which *LO* beats GM.

| Test                             | Base   | Class. | <i>IRLA</i> | <i>LO</i> <sub>1</sub> | Ridge | GM    | <i>LO</i> | $\Delta Q^2$ | >  |
|----------------------------------|--------|--------|-------------|------------------------|-------|-------|-----------|--------------|----|
| decanes ( $n = 34$ )             | -0.062 | 0.176  | 0.167       | 0.149                  | 0.860 | 0.514 | 0.406     | -0.072       | 15 |
| C6–C10, with $n_C$ ( $n = 100$ ) | 0.938  | 0.967  | 0.963       | 0.963                  | 0.991 | 0.977 | 0.977     | -0.000       | 25 |

Both pre-registered verdicts are negative. On the decane cohort the me-

dian  $\Delta Q^2 = -0.072$ , LO is ahead in 15 of 50 seeds, and the 95% interval is  $[-0.156, +0.001]$ . In the pooled size-adjusted set the values are  $-0.000$ , 25 of 50 and  $[-0.003, +0.002]$ . In the original box of half-width 2, reported as the pre-registration requires, the two tuned models are level on the decanes (medians 0.469 and 0.465). The Loyola model is marginally ahead on the pooled set (0.980 against 0.976). Therefore the decane verdict does not depend on the box. The lower decane  $Q^2$  of the hybrid control (0.406 against 0.465 in the narrow box) reflects the cost of a three-dimensional search over a wide box. It is not a penalty for  $\gamma$ . The pooled cohort was corrected after an external audit found one nonane’s second NIST entry skipped (Deviation D5 in the repository), and pooling grouped pages also moved one heptane value by  $0.2^\circ\text{C}$ . On the uncorrected cohort of 101 molecules the same test gave  $-0.000$ , 24 of 50,  $[-0.003, +0.002]$ , so the correction changes no conclusion. The intervals are molecule bootstraps conditional on the fitted predictions, as in Section 5.3. Ridge regression on the ten degree-pair counts reaches  $Q^2 = 0.860$  on the decanes, where every single index stays between 0.15 and 0.51, and 0.991 on the pooled set. A permutation check on the decanes (200 random permutations of the boiling points) gives it a median  $Q^2$  of  $-0.09$ , so the gap is not an artefact of the cross-validation procedure itself. A permutation test cannot, however, exclude every form of leakage. Branching effects on the boiling point are thus spread over several degree-pair counts in a way that no one-dimensional summary of the family, with or without  $\gamma$ , recovers. These conclusions concern acyclic alkanes and are the primary predictive evidence of this paper, while the vaporization-enthalpy analyses of Sections 5.2 and 5.3 are secondary diagnostics. Data, rule, exclusions and scripts are in the repository.

Data, scripts and verifiers that recompute every tabulated value are in the reproducibility package [23], where the nonane and boiling-point protocols were committed before the corresponding models were fitted. The termwise bounds, the convexity statement, the independence of the imbalance direction and the order-thirteen profile collision have also been machine-checked in Lean 4 with Mathlib. There graphs are modelled by their multisets of edge degree pairs, while equality characterisations were

not formalised. The scope of the formalisation is described in the Supplementary Material.

## 6 Conclusion

We extended the Gutman–Milovanović family by an exponential factor in the normalised degree imbalance of each edge and asked what this third coordinate adds. On the mathematical side we showed that the imbalance direction is affinely independent of the two parent directions once a vertex of degree three is present. The logarithm of the index is globally convex. The family factorises into a size coordinate and a shape coordinate. On the resulting ratio plane the pure-imbalance points discriminate exactly the histogram of edge imbalances. For trees of small order this histogram carries the full degree-pair information, so these points attain the discrimination limit of the whole degree-based class. From order thirteen onward (sixteen for chemical trees) they provably lose information, and we exhibited the colliding families. We also bounded the number of order exchanges between two chemical graphs along the imbalance axis. For each graph we characterised the parameter directions that leave the index unchanged up to scale.

On the chemometric side the evidence was negative. An apparent gain from the imbalance parameter on vaporization enthalpies replicated under a pre-registered protocol, but it disappeared once the parent family was searched over a box that did not truncate it. Pre-registered tests on measured boiling points of decanes and of a pooled set of C6–C10 alkanes showed no benefit from the third coordinate. There a linear model on the degree-pair counts outperformed every single index. The untuned pure-imbalance point is highly correlated with the parameter-free irregularity index and matched the tuned models on vaporization enthalpy. The compactness of the family therefore comes from linear imbalance weighting rather than from tuning. The third coordinate remains useful as an interpretable direction on the parent family along which the effect of degree imbalance can be analysed exactly. Open problems are listed in the Supplementary Material.

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